

Lecture 16

04/10/24 9:40 AM

* Cont. of Lecture 15 (APSP)

$d(i, j, l)$ = shortest path between i and j with at most l edges

$$d(i, j, l) = \min \begin{cases} d(i, j, l-1) \\ \min_{(u, j) \in E} \{ d(i, u, l-1) + w_{u, j} \} \end{cases}$$

3-dimensional array with $O(n^3)$ entries. $|V| = n$
Since each path has at most length $|V|$.

For each entry (i, j, l) , we do $(d_j + 1)$ lookups.

$d_j + 1 \leq n$
 $\therefore$ Worst case: $O(|V|^4)$

Consider $\min_{u \in V} \{ d(i, u, \frac{l}{2}) + d(u, j, \frac{l}{2}) \}$
shortest amongst all $i \rightsquigarrow u$ paths w/ at most $\frac{l}{2}$ edges
midpoint
 $i \text{ --- } u^* \text{ --- } j$
shortest amongst all $u \rightsquigarrow j$ paths w/ at most $\frac{l}{2}$ edges.

Obs: If π is a shortest path from i to j and u is on it,
then $\pi|_{i \rightsquigarrow u}$ is also the shortest path from i to u

PROOF: Suppose this is not true.

$\exists$ a path σ from i to u s.t. $d(\sigma|_{i \rightsquigarrow u}) < d(\pi|_{i \rightsquigarrow u})$

This contradicts that π is a shortest path from i to j

as we can construct a path $\sigma|_{i \rightsquigarrow u} \cdot \pi|_{u \rightsquigarrow j}$ which has shorter dist.

(Assume n is a power of 2)

$d(i, j, \frac{n}{2^k})$
 $\downarrow$
 n options each

$\Rightarrow$ 3-dim array has $n^2 \log n$ entries.
For each entry, we look at $2n$ entries.
 $\therefore O(n^3 \log n)$

BASE CASE: $d(i, u, 1) + d(u, j, 1)$

BOTTOM-UP

For k in $[1, \log n]$:

For i in V :

For j in V :

Min value = ∞

For u in V :

$val = d(i, u, 2^{k-1}) + d(u, j, 2^{k-1})$

If $val < \text{Min value}$:

Min value $\leftarrow val$

$d(i, j, 2^k) \leftarrow \text{Min value}$

TOP-DOWN

Shortest (i, j, l) :

For all u :

Shortest $(i, u, \frac{l}{2})$

Shortest $(u, j, \frac{l}{2})$

Recursive
Re-computing values
Wasteful

LONGEST INCREASING SUBSEQUENCE

$a_1 \ a_2 \ \dots \ a_n$ (distinct elements)

$i_1 < i_2 < \dots < i_k$ s.t. $a_{i_1} < a_{i_2} < a_{i_3} \dots < a_{i_k}$

e.g. 1 3 2 4 0 5 -2 $\rightarrow$ 1 2 4 5
1 3 4 5

$LIS([1, n])$ max

$LIS([1, n-1])$

$\hat{LIS}([1, n-1], a_n) + 1$

DOES NOT

CONTAIN a_n

= LIS s.t. all elements are smaller than a_n

CONTAINS a_n

$\hat{LIS}([1, i], x)$:

if $i = 0$:

return 0

$m = \hat{LIS}([1, i-1], x)$

if $a_i < x$:

$m \leftarrow \max \{ m, 1 + \hat{LIS}([1, i-1], a_i) \}$

return m

returning max. of
 $\{ \hat{LIS}([1, i-1], a_i) + 1, \hat{LIS}([1, i-1], x) \}$

$LIS([1, u])$:

return $\hat{LIS}([1, u], \infty)$

Obs: X takes only the values in the sequence + 1 more $\max(a_1, \dots, a_n) + 1$

Space: $O(n^2)$ entries and ≤ 2 lookups per entry

$\therefore$ Overall $O(n^2)$.

Brute force: 2^n subsequences